

Function Calling with Gemini & Building End-to-End GenAI Applications



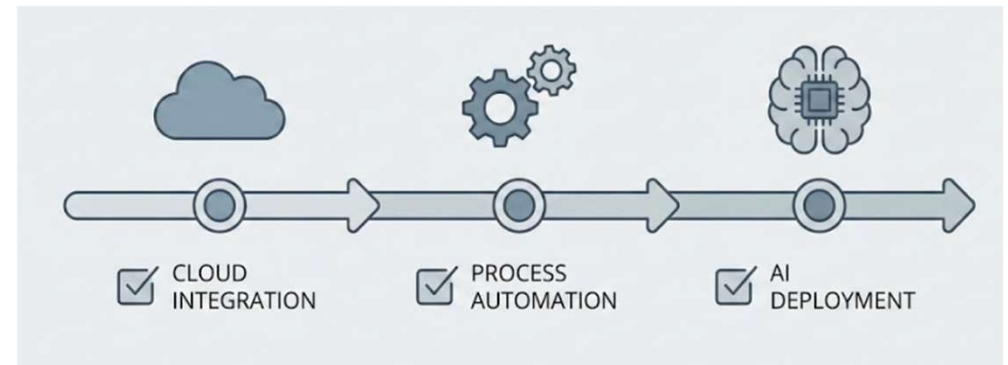
AI is transforming how modern applications are built and deployed. Gemini model introduces powerful capabilities that allow AI systems to move beyond conversation and interact directly with real-world tools and services. This session explores how Function Calling enables Gemini to connect with backend systems and how developers can build complete end-to-end Gen AI apps that solve practical business problems.

- Understanding Gemini AI
- Power of Function Calling
- Real-world applications
- Architecture and workflow
- Best practices



Agenda

- Introduction to Gemini
- Understanding Function Calling
- Importance in modern AI systems
- Workflow and architecture
- Real-world use cases
- Challenges and best practices



Introduction to Gemini



Gemini is Google's next-generation multimodal generative AI model designed to understand and process multiple forms of data such as text, images, code, and more. Unlike traditional AI systems, Gemini can reason across different data types, making it highly effective for building intelligent assistants, automation platforms, and decision-support systems.

- Advanced multimodal AI model
- Understands text, images, and code
- Built for real-world applications
- Supports tool and function calling
- Enables intelligent automation

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What is Function Calling?



It allows an AI model to interact with predefined software functions. Instead of only generating text, the AI can decide when a function is needed, extract the correct parameters, and request execution from the backend system. This transforms AI from a passive responder into an active problem-solving agent.

- Enables AI to call predefined functions
- AI determines when a function is required
- Sends structured arguments
- Backend executes the logic
- Returns real data to the AI



Why Do We Need Function Calling?



Large Language Models are powerful but limited when operating alone. They cannot directly access databases, perform transactions, or retrieve real-time information. Function Calling bridges this gap by securely connecting AI models with external systems, enabling accurate, controlled, and actionable responses.

- LLMs lack real-time system access
- Cannot execute business operations
- Prevents unreliable AI guesses
- Ensures controlled interactions
- Connects AI with enterprise systems

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Function Calling: Simple Explanation



It follows a structured process that begins with a user request and ends with an intelligent response powered by real data. The AI interprets the user's intent, selects the appropriate function, extracts parameters, and allows the backend to perform the required operation before delivering a natural language answer.

- User submits a request
- Gemini understands intent
- Selects the correct function
- Extracts parameters
- Backend executes logic
- AI returns the final response



User Request



Intent Analysis



Function Selection



Backend Execution



Final Response

Example Functions



Function Calling can support a wide variety of operations depending on business requirements. From retrieving weather updates to booking appointments, developers can define functions that allow AI to perform meaningful tasks across digital ecosystems.

- GetWeather(city)** – Fetch live weather data
- FetchUserDetails(user_id)** – Retrieve customer information
- BookAppointment(date, time)** – Automate scheduling
- GenerateInvoice(order_id)** – Create billing documents

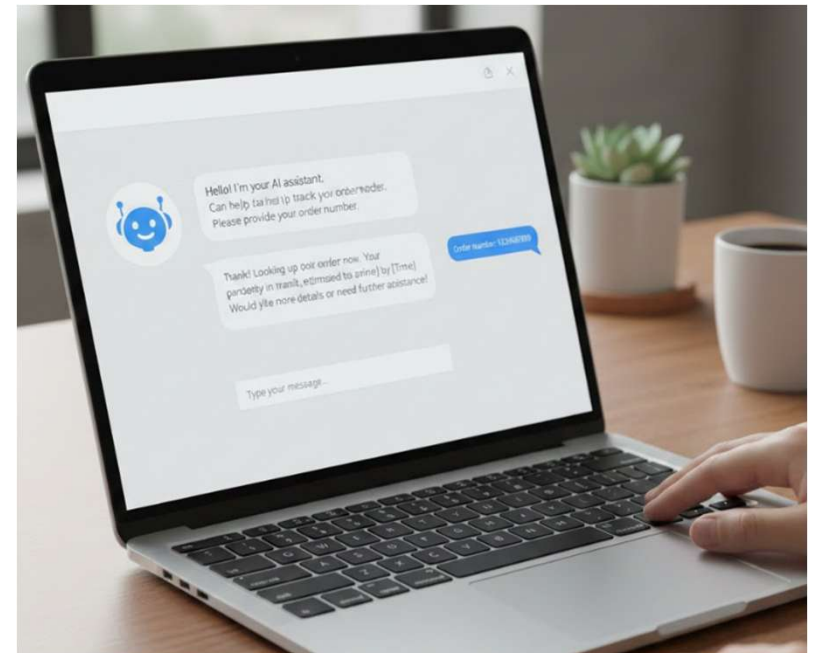


Real-Life Scenario: Customer Support



Function Calling significantly enhances customer service by enabling AI assistants to access order systems and provide accurate updates instantly. This reduces wait times, improves customer satisfaction, and lowers operational costs.

- User asks about order status
- Gemini calls OrderStatus()
- Backend retrieves tracking data
- AI explains status naturally



Real-Life Scenario: Healthcare



In healthcare support systems, AI can analyze symptoms and call specialized functions to assess risk levels. While it does not replace medical professionals, it can guide users toward informed decisions and timely care.

- User reports symptoms
- AI calls SymptomChecker()
- System evaluates risk
- AI provides safe guidance



What is an End-to-End GenAI Application?



An end-to-end Generative AI application combines multiple layers of technology to deliver intelligent user experiences. It integrates AI models with backend services, databases, and user interfaces to create scalable and production-ready systems.

- AI model at the core
- Backend services
- Database integration
- User interface
- Deployment and monitoring

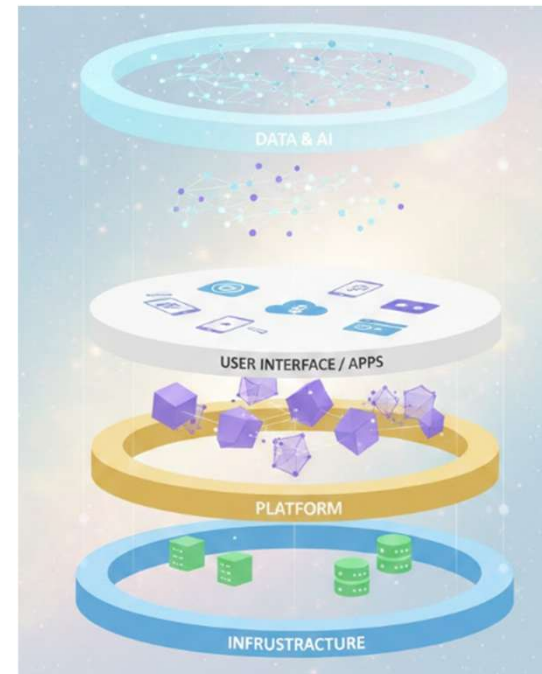


GenAI Application: Key Components



Each component in a GenAI application plays a critical role in ensuring performance, scalability, and reliability. When properly designed, these layers work together seamlessly to deliver fast and intelligent responses.

- Frontend (Web/Mobile)
- Backend APIs
- Gemini Model
- Function Calling Layer
- Database / External services

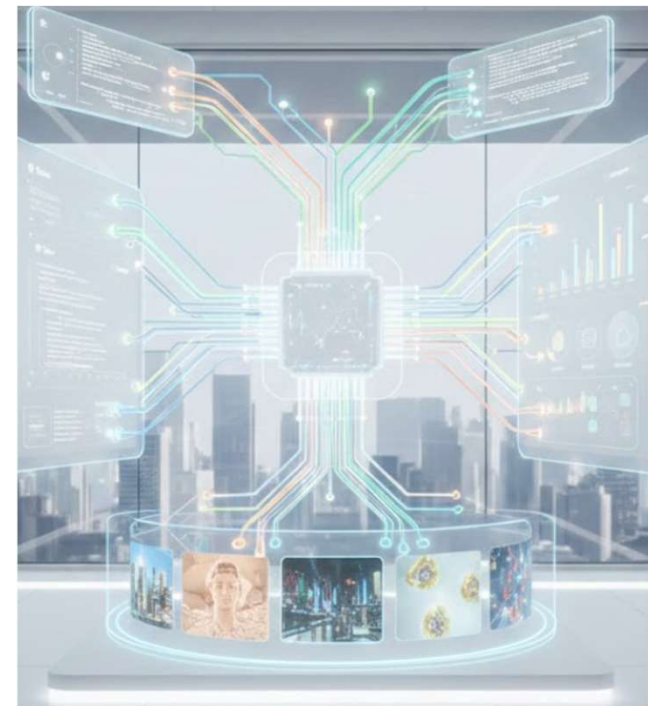


End-to-End Workflow



The workflow of a GenAI system ensures that every request is processed efficiently from user interaction to backend execution resulting in accurate and contextual responses.

- User sends request
- Frontend forwards it
- Gemini processes prompt
- Function triggers
- Backend executes
- Response returned

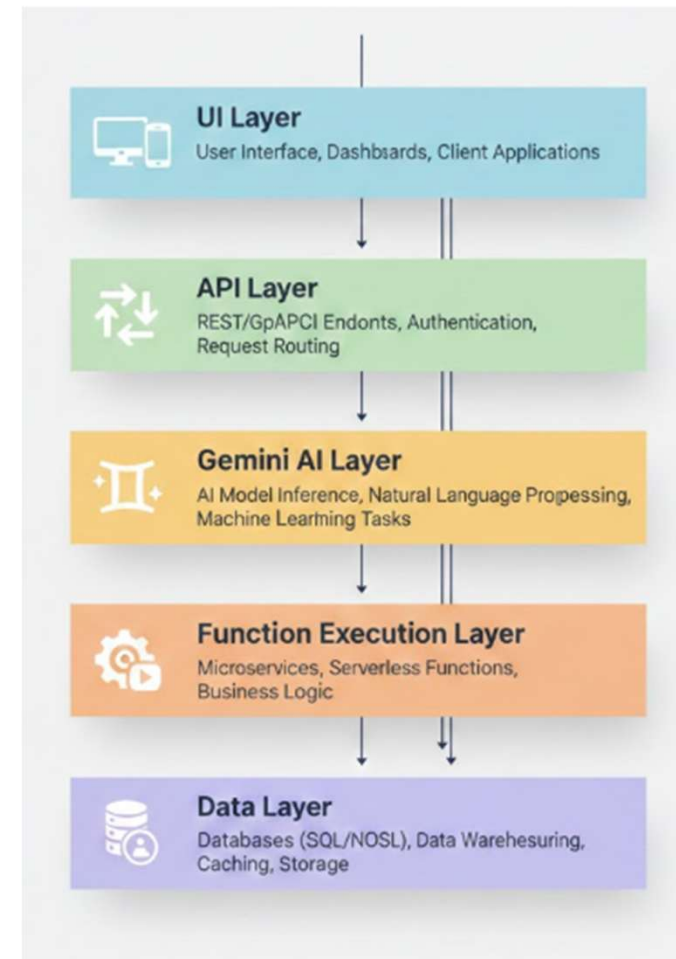


Architecture Diagram

A well-structured architecture is essential for building scalable AI applications. Separating responsibilities across layers improves maintainability, security, and performance.

- UI Layer
- API Layer
- Gemini AI Layer
- Function Execution Layer
- Data Layer

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Benefits of Function Calling



Function Calling elevates AI applications by improving accuracy and enabling real-world interaction. It minimizes hallucinations while ensuring that responses are grounded in actual data.

- Higher accuracy
- Real-time data access
- Secure execution
- Reduced hallucinations
- Better user trust



Challenges



Despite its advantages, implementing Function Calling requires careful planning. Developers must design reliable schemas, manage latency, and ensure strong security practices to prevent misuse.

- Poor schema design risks failure
- Error handling is critical
- Latency can impact UX
- Security vulnerabilities



Best Practices



Following proven best practices helps teams build robust and production-ready GenAI applications that remain secure, efficient, and scalable over time.

- Keep functions small and focused
- Validate inputs strictly
- Log all function calls
- Implement fallback responses
- Monitor performance

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